

# Standard Flips of Discrepancy One: Extremal $J$ -Normalization and the Meijer Aperture at $\nu = 1$

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August 2026

## Abstract

Shen and Shoemaker compute the extremal quantum spectrum of a standard flip  $X \dashrightarrow X'$  with exceptional locus  $\mathbf{P}(V) \subset X$ ,  $\text{rank } V = r$ ,  $\text{rank } V' = s$ , and show that the Gamma-class decomposition of  $H^*(X)$  attached to the Belmans–Fu–Raedschelders semiorthogonal decomposition is a decomposition into asymptotic classes. Their Theorem 4.4, which identifies an explicit hypergeometric series with the extremal  $J$ -function of the local model, assumes  $r - s > 1$ ; their Remark 4.5(3) asserts that for  $r - s \leq 1$  the series is not  $J$ -normalized; and the Barnes asymptotic expansion of their Section 7 is applied under the same inequality. Their Theorem 1.2 is printed only for  $r - s > 1$ , while their Theorems 1.4, 9.9 and 9.14 and their Corollary 1.5 are stated in a range that includes blow-ups. The printed proof chain for these later statements does not reach the discrepancy-one case  $r = s + 1$ ,  $s \geq 1$ , which contains every codimension-two blow-up.

We supply the two missing steps. First, the degree- $d$  summand of their formula (35) has  $z$ -order at most  $1 - s - (r - s)d$ ; for  $r = s + 1$  and  $s \geq 1$  this is at most  $-1$  for every  $d \geq 1$ , so the series is  $z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1})$ , with  $\mathbf{1}$  the unit class and  $\bar{\mathbf{t}} = t + \ln(q)c_1(T)$  the extremal parameter; no mirror-map correction arises, and uniqueness of the  $J$ -slice of Givental’s cone identifies the series with the extremal  $J$ -function. We prove the cone membership that this uses instead of quoting Remark 4.5(3) for it, whose printed attribution passes through a lemma of their Section 9 that presupposes Theorem 4.4: for split bundles it follows from Brown’s toric-fibration theorem together with the twisted theory of Coates and Givental, and the general case follows by a flag-bundle pullback and a deformation to the associated graded. None of these inputs restricts  $r - s$ . The only remaining formal failure of the normalization is the degenerate endpoint  $(r, s) = (1, 0)$ , whose point fibres contain no extremal line. Second, at  $\nu := r - s = 1$  the sector printed after their Lemma 7.4 is unavailable: their own Theorem A.1 is valid on  $|\arg t| < (\nu + \epsilon)\pi$  with  $\epsilon = 1$  only for  $\nu > 1$  and  $\epsilon = \frac{1}{2}$  at  $\nu = 1$ , and the correct  $\epsilon = \frac{1}{2}$  sector still meets the sector of their Proposition 8.2 in an open sector of opening  $2\pi$  that contains both the nonzero-eigenvalue ray and the tame ray. Once these repaired inputs are supplied, their Sections 9.1–9.4 impose no further restriction on the discrepancy, and Theorems 1.2, 1.4, 9.9 and 9.14 and Corollary 1.5 extend to every standard flip with  $r = s + 1$  and  $s \geq 1$ . No other part of the standard-flip argument is altered: the correction is confined to the  $J$ -normalization of the extremal hypergeometric series and to the Barnes aperture at  $\nu = 1$ .

**Keywords.** standard flip; quantum spectrum; Gamma class; asymptotic class; Meijer  $G$ -function; semiorthogonal decomposition.

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# 1 Introduction

A *standard flip*  $X \dashrightarrow X'$ , in the setting of Bondal–Orlov and of Belmans, Fu and Raedschelders [3], is built from the following data. There is a birational morphism  $p: X \rightarrow \bar{X}$  of smooth projective varieties, an isomorphism away from  $F = p^{-1}(Z)$  for a smooth  $Z \subset \bar{X}$ , and vector bundles  $V, V' \rightarrow Z$  of ranks  $r$  and  $s$  with

$$F \cong \mathbf{P}(V), \quad N_{F|X} \cong \pi^*(V') \otimes \mathcal{O}_{\mathbf{P}(V)}(-1),$$

where  $\pi: F \rightarrow Z$ . Blowing up  $F$  and contracting the exceptional divisor  $E \cong \mathbf{P}(V) \times_Z \mathbf{P}(V')$  in the other direction produces  $X'$ , which contains  $F' \cong \mathbf{P}(V')$  of codimension  $r$ . Throughout,  $r > s$ . Since  $F \subset X$  has codimension  $s$  and  $F' \subset X'$  has codimension  $r$ , the two blow-up formulas for the canonical class on  $\hat{X} = \text{Bl}_F X$  give  $\tau^* K_X - \tau'^* K_{X'} = (r - s)E$ . We call

$$\nu := r - s$$

the *discrepancy* of the flip;  $s = 1$  is the blow-up of  $\bar{X}$  along  $Z$ , and  $s = 0$  is the degenerate case  $X = F = \mathbf{P}(V)$ ,  $X' = \emptyset$ , of a projective bundle.

Shen and Shoemaker [13] compute the spectrum of quantum multiplication by  $c_1(X)$  after restricting the Novikov parameter to the extremal ray contracted by  $p$ , and prove that the Gamma-class decomposition of  $H^*(X)$  induced by the semiorthogonal decomposition of  $D^b(X)$  of [3] is a decomposition into strong asymptotic classes in the sense of Galkin–Golyshev–Iritani [7]. This is [13, Theorems 1.2, 1.4, 9.9 and 9.14, Corollary 1.5]; apart from [13, Theorem 1.2], which assumes  $r - s > 1$ , it is stated for standard flips with the explicit remark that it covers the blow-up case  $s = 1$  and the projective-bundle case  $s = 0$ .

The proof passes through an explicit hypergeometric series. In the local model  $T = \text{tot}(\pi^* V' \otimes \mathcal{O}_{\mathbf{P}(V)}(-1))$ , after the extremal specialization [13, (33)] of Novikov and parameter variables, [13, Theorem 4.4] states that the extremal  $J$ -function of  $T$  is

$$J^T(q, t, z) = z e^{t/z} q^{c_1(T)/z} \sum_{d \geq 0} q^{(r-s)d} \frac{\prod_{j=1}^s \prod_{m=0}^{d-1} (\sigma_j - H - mz)}{\prod_{i=1}^r \prod_{m=1}^d (\rho_i + H + mz)}, \quad (1)$$

which is [13, (35)]. Everything after Section 4 of [13] runs on this formula: the ring presentation [13, (37)] is deduced from it, the extremal spectrum is read off from that presentation, and the modified  $J$ -function of [13, (39)–(40)] used in the asymptotic analysis of Sections 7–9 is defined from it.

Three hypotheses in the printed argument constrain  $\nu$ . [13, Theorem 4.4] is stated under  $r - s > 1$ . [13, Remark 4.5(3)] asserts that for  $r - s \leq 1$  the right-hand side of (1) is not of the form  $z\mathbf{1} + (\text{lower order in } z)$  and therefore cannot equal the  $J$ -function of  $T$ , although it does lie on Givental’s Lagrangian cone of  $T$  and so is an  $I$ -function. Finally, the Barnes asymptotic expansion of the Meijer  $G$ -function of [13, Lemma 7.4] is introduced by the clause “For  $r - s > 1$ ”, and the sector [13, (64)] printed with it is derived under that inequality. Under the standing assumption  $r > s$ , the range excluded by these three hypotheses is exactly  $\nu = 1$ , that is

$$r = s + 1.$$

It contains every codimension-two blow-up, the case  $(r, s) = (2, 1)$ . Thus [13, Theorem 1.2] is proved only in its stated range  $r - s > 1$ , while [13, Theorems 1.4, 9.9 and 9.14] and [13, Corollary 1.5], which are stated for standard flips and blow-ups without a restriction on  $\nu$ , rely on inputs whose printed proofs do not cover  $r - s = 1$ .

This note supplies the two missing steps. The second uses only [13, Theorems A.1 and A.2], in the form the authors state them. The first uses (1) together with the fact that

it lies on Givental's Lagrangian cone. That fact is asserted in [13, Remark 4.5(3)], the same remark whose other half we contradict, so we do not quote it: Section 3 derives it from the sources [4] and [5] that the authors name, and checks that none of them restricts  $r - s$ .

**Theorem 1.1.** *Let  $r = s + 1$  with  $s \geq 1$ . Then, in the extremal specialization [13, (33)] and with the same formal variables and cohomological completion as in [13, Theorem 4.4], the degree- $d$  summand of (1) has  $z$ -order at most  $-1$  for every  $d \geq 1$ . Consequently (1) equals  $z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1})$  with  $\bar{\mathbf{t}} = t + \ln(q)c_1(T)$ , and equals the extremal  $J$ -function of  $T$ . In particular the presentation [13, (37)] of [13, Theorem 4.6] holds for  $r = s + 1$ ,  $s \geq 1$ , as does [13, Theorem 1.2], with the single nonzero eigenvalue  $\lambda_0(q) = (-1)^s q$  of multiplicity  $\text{rank } H^*(Z)$ .*

The negative assertion of [13, Remark 4.5(3)] is therefore accurate, in the range  $r > s$  it addresses, only for  $s = 0$ , where  $(r, s) = (1, 0)$  and the degree-one summand of (1) does contribute in  $z$ -order zero.

**Theorem 1.2.** *Let  $r = s + 1$  with  $s \geq 1$ , and let  $k \in \{0, 1\}$ . The asymptotic expansion of [13, Proposition 7.5] for the component indexed by  $m = -k$  is valid on  $|\arg(z/q) + (2m + s)\pi| < \frac{3\pi}{2}$ ; [13, Theorem A.1] does not supply it on the wider sector obtained by setting  $r - s = 1$  in [13, (64)]. The expansion of [13, Proposition 8.2] is valid on  $|\arg(z/q) - (1 - s)\pi| < \frac{3\pi}{2}$ . These two sectors meet in an open sector of opening  $2\pi$  containing both the eigenvalue ray  $\arg(z/q) = -(2m + s)\pi$  and the tame ray  $\arg(z/q) = (1 - s)\pi$ .*

For a codimension-two blow-up,  $(r, s) = (2, 1)$  and  $k = m = 0$ , the two sectors are  $-\frac{5\pi}{2} < \arg(z/q) < \frac{\pi}{2}$  and  $|\arg(z/q)| < \frac{3\pi}{2}$ , and they meet in  $-\frac{3\pi}{2} < \arg(z/q) < \frac{\pi}{2}$ , which contains the tame ray 0 and the eigenvalue ray  $-\pi$ .

**Corollary 1.3.** *[13, Theorems 1.2, 1.4, 9.9 and 9.14] and [13, Corollary 1.5] hold for every standard flip with  $r = s + 1$  and  $s \geq 1$ , in particular for every codimension-two blow-up.*

The mechanism behind Theorem 1.1 is a single count, and it is worth saying where the natural first count goes wrong. The numerator of the degree- $d$  summand of (1) has  $sd$  factors and the denominator has  $rd$  factors. Treating all of them as linear in  $z$  gives  $1 + sd - rd = 1 - \nu d$  for the maximal  $z$ -order, which is 0 at  $d = 1$  when  $\nu = 1$ : a curve-induced contribution in  $z$ -order zero, hence a nontrivial mirror map, hence exactly the failure asserted in [13, Remark 4.5(3)]. But the inner index in the numerator runs over  $0 \leq m \leq d - 1$ , so  $s$  of its factors are  $\sigma_j - H$ , free of  $z$ . The correct maximal order is  $1 + s(d - 1) - rd = 1 - s - \nu d$ , and at  $\nu = 1$  the  $s$  missing powers of  $z$  are precisely what pushes every  $d \geq 1$  summand below  $z^0$  as soon as  $s \geq 1$ . Once no  $z$ -order-zero contribution survives, the  $z^0$  coefficient of (1) is the parameter  $\bar{\mathbf{t}}$  itself, the mirror map is the identity, and uniqueness of the  $J$ -slice of the cone converts cone membership into an identification.

Theorem 1.2 is not a strengthening but a correction of an aperture. The Meijer  $G$ -function of [13, Lemma 7.4] has  $\nu = r - s$ , and [13, Theorem A.1] — Barnes' expansion in the form of Meijer [10] and Luke [9] — is valid on  $|\arg t| < (\nu + \epsilon)\pi$  with  $\epsilon = 1$  for  $\nu > 1$  but only  $\epsilon = \frac{1}{2}$  for  $\nu = 1$ . The sector [13, (64)] is the  $\epsilon = 1$  sector; substituting  $r - s = 1$  into it claims an aperture  $\pi/2$  wider on each side than Barnes' expansion supports. The point of Theorem 1.2 is that the smaller, correct sector is still large enough: the common sector with [13, Proposition 8.2] retains opening  $2\pi$ , so the oriented comparison of the nonzero-eigenvalue and tame expansions that the argument of [13, Section 9] performs is available at  $\nu = 1$  as it is for  $\nu > 1$ .

What is not covered is the formal endpoint  $(r, s) = (1, 0)$ , the rank-one projective bundle, where  $\mathbf{P}(V) = Z$ ,  $X' = \emptyset$ , and the point fibres contain no line class  $[L]$ . Its displayed

hypergeometric expression also fails the required order count. Nothing here bears on  $\nu > 1$ , where the argument of [13] already applies, and nothing here changes any statement of [13] outside the two hypotheses named above.

Section 2 fixes notation and records the three places where  $\nu$  is constrained in [13], together with what depends on each. Section 3 proves Theorem 1.1, Section 4 proves Theorem 1.2, and Section 5 assembles Corollary 1.3 and states the boundary of the repair.

*Convention.* Numbered items of [13] are always cited explicitly, as in [13, Theorem 4.4] or [13, (35)], and refer to the version arXiv:2502.08762v2. Numbered items of [5], and the discussion “On Serre duality” cited in Section 3, refer to arXiv:math/0110142v2. Displays numbered here, such as (1), are our own.

## 2 The extremal setting and the three constraints on $\nu$

We keep the notation of [13]. Let  $H = c_1(\mathcal{O}_{\mathbf{P}(V)}(1))$ , let  $\rho_1, \dots, \rho_r$  and  $\sigma_1, \dots, \sigma_s$  be the Chern roots of  $V$  and  $V'$  pulled back to  $\mathbf{P}(V)$ , and let  $T = \text{tot}(\pi^*V' \otimes \mathcal{O}_{\mathbf{P}(V)}(-1))$  be the local model, so that  $c_1(T) = (r - s)H + c_1(Z) + c_1(V) + c_1(V')$  by [13, (32)].

The  $J$ -function of a smooth projective  $Y$  is  $J^Y(Q, t, z) = zS^Y(Q, t, z)^{-1}\mathbf{1}$  as in [13, (20)]; it satisfies

$$J^Y(Q, t, z) = z\mathbf{1} + t + O(z^{-1}). \quad (2)$$

The *extremal* specialization [13, (28) and (33)] sets  $Q^d = 1$  for  $d = k[L]$  with  $[L]$  the class of a line in a fibre of  $\mathbf{P}(V) \subset T \rightarrow Z$ , sets  $Q^d = 0$  for all other  $d$ , and replaces the parameter by

$$\bar{t} = t + \ln(q)c_1(T), \quad t \in H^*(Z). \quad (3)$$

In the specialized variables (2) reads  $J^T(q, t, z) = z\mathbf{1} + \bar{t} + O(z^{-1})$ , and the prefactor  $ze^{t/z}q^{c_1(T)/z} = ze^{\bar{t}/z}$  of (1) is exactly the  $d = 0$  summand. Whether (1) is the  $J$ -function is therefore the question of whether the summands with  $d \geq 1$  contribute in  $z$ -order  $\geq 0$ .

Three places in [13] carry a hypothesis on  $\nu = r - s$ , and the whole of Sections 5–9 depends on the first and third through the chain recorded below. We list each with the inequality it states and the statements that use it.

| Location in [13]       | Hypothesis     | Used by                                   |
|------------------------|----------------|-------------------------------------------|
| Theorem 4.4            | $r - s > 1$    | (37); Theorem 4.6; (39)–(40)              |
| Remark 4.5(3)          | $r - s \leq 1$ | nothing; a negative assertion             |
| Section 7, before (64) | $r - s > 1$    | (64); Proposition 7.5 and its corollaries |

The first row is the load-bearing one. [13, Theorem 4.6] deduces the presentation

$$QH_{\text{ext}}^*(T) = H^*(Z)[H][[q]] / \left\langle \prod_{i=1}^r (\rho_i + H) = q^{r-s} \prod_{j=1}^s (\sigma_j - H) \right\rangle, \quad (4)$$

which is [13, (37)], from (1) by exhibiting a quantum differential operator annihilating  $J^T$  and quoting [6, Theorem 10.3.1] and [12, Lemma 2]. That argument uses  $r - s > 1$  only through [13, Theorem 4.4]. The spectral computations [13, Propositions 5.1 and 5.2, Corollary 5.3] then use (4) and the ring homomorphism of [13, Theorem 3.3], and carry no hypothesis on  $\nu$  of their own; the inequality in [13, Theorem 1.2] is inherited from [13, Theorem 4.4]. Likewise, the modified extremal  $J$ -function [13, (40)] used throughout Sections 7–9 is derived from (1), and [13, Proposition 9.1] uses (4) again. Establishing (1) at  $\nu = 1$  therefore unblocks all of this at once.

The second row asserts the opposite of what we prove, and nothing in [13] uses it. The remark then adds that (1) nevertheless lies on Givental’s overruled Lagrangian cone of  $T$

and is therefore an  $I$ -function; it is this second, positive assertion that the authors explicitly say they will not need, and the same holds for its counterpart [13, Remark 4.7] for  $T'$ . Cone membership is what we use, and Section 3 proves it rather than quoting the remark. The authors attribute it to [4], [5] and [13, Lemma 9.6]; the first two carry no hypothesis on  $\nu$ , while [13, Lemma 9.6] is not available for this purpose at  $\nu = 1$ , for the reason recorded in Remark 3.8.

The third row governs the aperture of the Barnes expansion only. Nothing in Sections 8 and 9 of [13] constrains  $\nu$ : the sector [13, (78)] of [13, Proposition 8.2] comes from [13, Theorem A.2], whose hypothesis  $|\arg t| < (\frac{\nu}{2} + 1)\pi$  carries no  $\epsilon$  and is valid at  $\nu = 1$ ; and the reductions of [13, Sections 9.1–9.4] — weak-to-strong in the local model, the reduction to the split case, the cohomological reduction to the local model, and the comparison of the Fourier–Mukai transform with the local model — state no inequality beyond  $r > s$ , although the reduction to the split case runs through [13, Lemmas 9.4–9.6], whose proofs read (1) as the extremal  $J$ -function; see Remark 3.8.

### 3 $J$ -normalization at discrepancy one

Let  $\phi: Y \rightarrow Z$  be the flag bundle of  $V \oplus V'$ , so that  $\phi^*V$  and  $\phi^*V'$  carry filtrations whose successive quotients are line bundles, and  $\phi^*: H^*(Z) \rightarrow H^*(Y)$  is injective; this is the splitting principle for extremal quantum cohomology, [13, Corollary 4.3]. The Chern roots  $\rho_i$  and  $\sigma_j$  exist individually on  $Y$ , and we continue to write  $V, V', \rho_i, \sigma_j$  for the pullbacks. Write

$$A = H^*(\mathbf{P}_Y(V)) \cong H^*(Y)[H] / \langle \prod_{i=1}^r (\rho_i + H) \rangle$$

for the coefficient ring of (1) after this pullback. It is a finite-dimensional graded  $\mathbf{C}$ -algebra whose elements of positive degree are nilpotent. Each factor of the denominator of (1) has the form  $mz + a$  with  $m \geq 1$  and  $a = \rho_i + H \in A$ , so

$$\frac{1}{mz + a} = \frac{1}{mz} \sum_{l \geq 0} \left( \frac{-a}{mz} \right)^l \in z^{-1} A[z^{-1}]$$

is a finite sum, so each summand of (1) lies in  $zA[z^{-1}][[q]]$  once the  $\ln q$  carried by the prefactor is handled as described before Lemma 3.4. By the  $z$ -order of such an element we mean the largest  $n$  with a nonzero  $z^n$  coefficient. A  $z$ -order bound proved after the pullback is a coefficientwise statement about classes in the image of an injective map, so it descends to the original cohomology ring; the bound below is in any case stated in terms of  $r, s$  and  $d$  alone.

**Lemma 3.1.** *For every  $d \geq 1$ , the degree- $d$  summand of (1) has  $z$ -order at most*

$$1 + s(d - 1) - rd = 1 - s - \nu d.$$

*Proof.* The numerator  $\prod_{j=1}^s \prod_{m=0}^{d-1} (\sigma_j - H - mz)$  has  $sd$  factors. The  $s$  factors with  $m = 0$  are  $\sigma_j - H$  and contain no  $z$ ; each of the remaining  $s(d - 1)$  factors is linear in  $z$ . So the numerator is a polynomial in  $z$  over  $A$  of degree at most  $s(d - 1)$ . The denominator  $\prod_{i=1}^r \prod_{m=1}^d (\rho_i + H + mz)$  has  $rd$  factors, all with  $m \geq 1$ , so by the expansion above its inverse lies in  $z^{-rd} A[z^{-1}]$ . Multiplying by the leading factor  $z$  of (1) adds one.  $\square$

The two hypotheses of Theorem 1.1 enter here and nowhere else.

**Corollary 3.2.** *If  $\nu = 1$  and  $s \geq 1$ , then every summand of (1) with  $d \geq 1$  has  $z$ -order at most  $1 - s - d \leq -1$ , and hence*

$$z e^{t/z} q^{c_1(T)/z} \sum_{d \geq 0} q^{(r-s)d} \frac{\prod_{j=1}^s \prod_{m=0}^{d-1} (\sigma_j - H - mz)}{\prod_{i=1}^r \prod_{m=1}^d (\rho_i + H + mz)} = z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1}),$$

with  $\bar{\mathbf{t}} = t + \ln(q)c_1(T)$  as in (3).

*Proof.* The bound of Lemma 3.1 is  $1 - s - d$ , which is at most  $-1$  for  $d \geq 1$  and  $s \geq 1$ . The  $d = 0$  summand is  $z e^{t/z} q^{c_1(T)/z} = z e^{\bar{\mathbf{t}}/z} = z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1})$ .  $\square$

*Remark 3.3.* Both hypotheses are sharp. For  $\nu = 1$  and  $s = 0$ , so  $(r, s) = (1, 0)$ , the bound of Lemma 3.1 is  $1 - d$  and is attained by the formal  $d = 1$  term of the hypergeometric expression:

$$z \cdot \frac{q}{\rho_1 + H + z} = q\mathbf{1} - \frac{q(\rho_1 + H)}{z} + \dots$$

Its  $z^0$  coefficient is  $q\mathbf{1}$ , so this formal expression is not  $J$ -normalized. Geometrically  $\mathbf{P}(V) = Z$  has point fibres and no line class  $[L]$ , so  $(1, 0)$  is a degenerate endpoint of the extremal-line setup, not a further standard-flip case repaired here. For  $\nu = 0$  the bound is the  $d$ -independent value  $1 - s$  and the sum is not  $q$ -adically convergent, consistently with the standing assumption  $r > s$  in [13].

To convert the normalization into an identification we use the standard characterization of the  $J$ -function as a slice of Givental's cone. Let  $\mathcal{L}_T$  denote the overruled Lagrangian cone of the genus-zero Gromov–Witten theory of  $T$ , in the extremal specialization (3).

The parameter  $\bar{\mathbf{t}} = t + \ln(q)c_1(T)$  is not literally defined over  $\mathbf{C}[[q]]$ . Two readings remove the difficulty, and they agree. One may adjoin  $\ln q$  to the coefficients, the standard extension used with the divisor equation, over which both the extremal cone and (1) are defined; or one may factor off the prefactor  $q^{c_1(T)/z}$  and work with the remaining series, in which  $q$  appears only in integral powers. In either reading the degree- $d$  summand carries  $q^{(r-s)d}$  by (1). That weight is positive because  $r > s$ , so the series converges  $q$ -adically and (1) is a point of the ambient symplectic space.

**Lemma 3.4** ( $J$ -slice uniqueness; [8], [5]). *Let  $Y$  be  $T$  or  $\mathbf{P}(V)$  in the extremal specialization, and let  $f \in \mathcal{L}_Y$  be of the form  $f = z\mathbf{1} + \tau + O(z^{-1})$  for some  $\tau$  in the parameter space. Then  $f = J^Y(\tau, z)$ .*

This is the mechanism behind every mirror theorem of Givental type: a point of the cone with a prescribed  $z^0$  coefficient is the  $J$ -function at that coefficient, and the mirror map is precisely the difference between the prescribed  $z^0$  coefficient and the parameter one started with.

The second ingredient is that (1) lies on  $\mathcal{L}_T$  at all. This is asserted in [13, Remark 4.5(3)], the same remark whose other assertion Corollary 3.2 contradicts, so we prove it instead of quoting it. We begin with split bundles, where the two sources the authors name do the work.

**Lemma 3.5.** *Let  $r > s \geq 0$  with  $r \geq 2$ , and suppose  $V = \bigoplus_{i=1}^r L_i$  and  $V' = \bigoplus_{j=1}^s M_j$ . Then the right-hand side of (1) lies on  $\mathcal{L}_T$ .*

*Proof.* The projective bundle  $\mathbf{P}(V) \rightarrow Z$  is a toric fibration with fibre  $\mathbf{P}^{r-1}$ . Brown's theorem [4, Theorem 1] states that the hypergeometric modification of the  $J$ -function of the base lies on the overruled cone of the total space, and [4, Corollary 1] is the projective-fibration case. Setting the Novikov variables of  $Z$  to zero and keeping the fibrewise variable

is the extremal specialization [13, (33)], a homomorphism of Novikov rings by [13, Proposition 4.1]; it leaves [13, (34)], which therefore lies on the extremal cone  $\mathcal{L}_{\mathbf{P}(V)}$  of  $\mathbf{P}(V)$ . Brown's hypotheses concern the fibration  $\mathbf{P}(V) \rightarrow Z$  and the Chern roots  $\rho_1, \dots, \rho_r$ ; no second bundle occurs in them, so they cannot constrain  $r - s$ .

Now the twist. A stable map  $f: C \rightarrow \mathbf{P}(V)$  of class  $k[L]$  with  $k \geq 1$  has image in a fibre of  $\mathbf{P}(V) \rightarrow Z$  by [13, Lemma 2.6]. There  $V'(-1) = \pi^*V' \otimes \mathcal{O}_{\mathbf{P}(V)}(-1)$  pulls back to a sum of  $s$  line bundles of degree  $-k_i$  on each component  $C_i$ , where  $k_i \geq 0$  is the degree of  $f|_{C_i}$  and  $\sum_i k_i = k$ . A section vanishes on every component with  $k_i > 0$ , and, the dual graph being a connected tree with at least one such component, on the contracted components as well; so  $H^0(C, f^*V'(-1)) = 0$  and  $V'(-1)$  is concave along every extremal class, while its dual  $W := (V'(-1))^\vee = \pi^*(V')^\vee \otimes \mathcal{O}_{\mathbf{P}(V)}(1)$  is convex there. Twisting by the inverse  $\mathbf{C}^\times$ -equivariant Euler class for the fibrewise action, whose parameter we write  $\lambda$ , gives the Gromov–Witten theory of the total space of a concave bundle, and the hypergeometric modification [5, Theorem 2] of the  $J$ -function has a Serre-dual partner obtained by replacing the Euler class of  $W$  with the inverse equivariant Euler class of  $W^\vee$ ; that partner is stated in the discussion “On Serre duality” of [5].

The dual modification is termwise, so its degree- $d$  factor can be read off. The Chern roots of  $W$  are  $H - \sigma_j$ , and each pairs to  $d$  on  $d[L]$ , so the factor contributed by the  $j$ -th root is  $\prod_{l=-d+1}^0 (-\lambda - (H - \sigma_j) + lz)$ , which is empty for  $d = 0$ . In positive degree the concave local theory admits the non-equivariant specialization, and letting  $\lambda \rightarrow 0$  and writing  $m = -l$  turns this into  $\prod_{m=0}^{d-1} (\sigma_j - H - mz)$ .

The modification applies to a  $J$ -function, so we record that [13, (34)] is one. For  $r \geq 2$  the count of Lemma 3.1, applied with  $s = 0$ , gives its degree- $d$  summand  $z$ -order at most  $1 - rd \leq -1$  for every  $d \geq 1$ ; so [13, (34)] is  $J$ -normalized and, lying on  $\mathcal{L}_{\mathbf{P}(V)}$ , equals  $J^{\mathbf{P}(V)}$  by Lemma 3.4. Applying the modification to it, and reparametrizing the extremal variable as in [13, (33)], so that the class  $d[L]$  carries the weight  $q^{\langle c_1(T), d[L] \rangle} = q^{(r-s)d}$  and the prefactor becomes  $q^{c_1(T)/z}$ , gives (1). The modification factor carries no  $q$ ; the exponent changes only because the reparametrization is by  $c_1(T) = c_1(\mathbf{P}(V)) + c_1(V'(-1))$ . The  $d = 0$  summand is untouched by the modification and remains the classical term  $ze^{\tilde{t}/z}$ . The modification involves the Chern roots of  $V'(-1)$  and nothing else; here too  $r$  and  $s$  enter only as numbers of factors.  $\square$

The passage from split bundles to general ones is where the printed attribution of [13, Remark 4.5(3)] cannot be followed at  $\nu = 1$ , so we spell it out. What it transports is the identity between (1) and  $J^T$ , not membership in an infinite-dimensional cone.

**Lemma 3.6.** *Let  $r > s \geq 0$ . If the right-hand side of (1) equals the extremal  $J$ -function for every local model whose bundles split, then it does so for every local model.*

*Proof.* Let  $\phi: Y \rightarrow Z$  be the flag bundle of  $V \oplus V'$  and  $\tilde{\phi}: T_Y = Y \times_Z T \rightarrow T$  the induced map. The Cartesian square identifies the relative obstruction theories and hence the virtual classes of  $\overline{M}_{0,n}(T_Y, d)$  and  $\overline{M}_{0,n}(T, d)$  for  $d$  in the span of  $[L]$ ; this is the content of [13, Theorem 4.2] and, for the fundamental solution, of [13, Lemma 9.5]. Applied to the two-point descendant invariants of [13, (18)] it gives  $\tilde{\phi}^* S^T = S^{T_Y} \tilde{\phi}^*$  and therefore  $\tilde{\phi}^* J^T = J^{T_Y}$ . Both sides of (1) are symmetric in the Chern roots [13, Remark 4.5(2)], and the expression for  $T_Y$  is the pullback of the expression for  $T$ . Since  $\tilde{\phi}^*$  is injective by Leray–Hirsch [13, Corollary 4.3], it suffices to prove the identity on  $T_Y$ , where  $V$  and  $V'$  are filtered with line-bundle quotients.

It remains to replace a filtered bundle by its associated graded. Take one step  $0 \rightarrow A \rightarrow V \rightarrow Q \rightarrow 0$  of the filtration; in this paragraph  $A$ ,  $Q$  and  $t$  are the sub- and quotient bundles and the coordinate on  $\mathbf{A}^1$ , not the coefficient ring above, the Novikov variable of (2) or the

parameter of (3). The construction of [13, Lemma 9.4] produces a bundle  $\ker(g)$  on  $Z \times \mathbf{A}^1$  with  $\ker(g)_1 = V$  and  $\ker(g)_0 = A \oplus Q$ , hence a smooth projective family  $\mathbf{P}(\ker g) \rightarrow Z \times \mathbf{A}^1$  carrying the bundle  $V'(-1)$ . For  $k \geq 1$ , the argument of [13, Proposition 3.1], applied to the projective bundle  $\mathbf{P}(\ker g) \rightarrow Z \times \mathbf{A}^1$ , makes the moduli spaces  $\overline{M}_{0,n}(\mathbf{P}(\ker g)_t, k[L])$  a smooth proper family over  $\mathbf{A}^1$ , and concavity makes  $R^1\pi_{\mathcal{C}*}f^*V'(-1)$ , for  $\pi_{\mathcal{C}}$  and  $f$  the universal curve and universal map, a relative locally free obstruction bundle of rank  $s(k-1)$ . Its Euler class, capped with the relative fundamental class, gives a relative cycle whose fibrewise restrictions are the extremal virtual classes of [13, Corollary 3.2]. The relative evaluation maps and  $\psi$ -classes define relative descendant cycles, and pushing these to  $\mathbf{A}^1$  gives classes in  $H^0(\mathbf{A}^1)$ , hence constants. So the extremal genus-zero descendant invariants, and with them the extremal  $J$ -functions, agree under the isomorphism of cohomology rings of [13, Lemma 9.4]. The right-hand side of (1) is likewise unchanged, because  $V$  and  $A \oplus Q$  have the same Chern roots. Repeating the construction for the filtration of  $V'$ , and iterating both filtrations as in [13, Lemma 9.6], reaches the split associated-graded local model.  $\square$

**Proposition 3.7.** *Let  $r = s + 1$  with  $s \geq 1$ . Then the right-hand side of (1) equals the extremal  $J$ -function  $J^T(q, t, z)$  of the local model, in the variables and completion of [13, Theorem 4.4].*

*Proof.* Assume first that  $V$  and  $V'$  split. By Lemma 3.5 the right-hand side of (1) is a point of  $\mathcal{L}_T$ , and by Corollary 3.2 its expansion is  $z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1})$ . So it is a point of  $\mathcal{L}_T$  with prescribed  $z^0$  coefficient  $\bar{\mathbf{t}}$ , and by uniqueness of the  $J$ -slice, Lemma 3.4 with  $\tau = \bar{\mathbf{t}}$ , it is  $J^T(q, t, z)$ . The general case follows from Lemma 3.6.  $\square$

*Remark 3.8.* The route just taken differs from the printed attribution of [13, Remark 4.5(3)] in one place. The authors deduce cone membership from [4], [5] and [13, Lemma 9.6]. The first two are the split-case inputs of Lemma 3.5 and carry no hypothesis on  $\nu$ . But [13, Lemma 9.6] rests on [13, Lemma 9.4], whose identification of the fundamental solutions of  $T$  and its associated-graded model is deduced from (1) read as a  $J$ -function formula, that is, from [13, Theorem 4.4] with its hypothesis  $r - s > 1$ . At  $\nu = 1$  that formula is what is being established, so Lemma 3.6 appeals to the underlying deformation instead of to the conclusion drawn from it. Once Proposition 3.7 is available, [13, Lemmas 9.4–9.6] hold at  $\nu = 1$  as printed, their input having been supplied; this is what Section 5 uses.

**Corollary 3.9.** *Let  $r = s + 1$  with  $s \geq 1$ . Then the presentation (4) of [13, Theorem 4.6] holds, the modified extremal  $J$ -function [13, (39)–(40)] of  $T$  is as printed, and [13, Theorem 1.2] holds: after the extremal specialization,  $c_1(X) \star_q -$  on  $QH^*(X)$  has eigenvalue 0 with multiplicity  $\text{rank } H^*(X')$  and the single nonzero eigenvalue*

$$\lambda_0(q) = e^{-\pi\sqrt{-1}s}q = (-1)^sq$$

*with multiplicity  $\text{rank } H^*(Z)$ .*

*Proof.* The proof of [13, Theorem 4.6] specializes the parameter  $t$  of [13, Theorem 4.4], exhibits a differential operator annihilating  $J^T$ , and concludes with [6, Theorem 10.3.1] and [12, Lemma 2]; it uses  $r - s > 1$  only to know that (1) is  $J^T$ . Proposition 3.7 supplies that, so the proof applies verbatim, and [13, (39)–(40)] follow by the same substitution as in [13, Section 4.3].

For the spectrum, [13, Propositions 5.1 and 5.2] concern the abstract ring (4) and its subspaces and assume only  $r > s$ , and [13, Corollary 5.3] combines them with the ring homomorphism  $i^*: QH_{\text{ext}}^*(X) \rightarrow QH_{\text{ext}}^*(T)$  of [13, Theorem 3.3]. Once (4) is available at  $\nu = 1$ , that argument is unchanged. Its output is  $r - s$  nonzero eigenvalues  $\lambda_k(q) =$



$(r-s)e^{-\pi\sqrt{-1}(2k+s)/(r-s)}q$  for  $0 \leq k < r-s$ ; at  $r-s=1$  only  $k=0$  occurs, and  $\lambda_0(q) = e^{-\pi\sqrt{-1}s}q$ .  $\square$

*Remark 3.10.* The companion statement [13, (34)] for  $\mathbf{P}(V)$  is the  $s=0$  case of the same expression, and Lemma 3.1 applied to it gives maximal  $z$ -order  $1-rd$ . This is at most  $-1$  for every  $d \geq 1$  as soon as  $r \geq 2$ , which holds whenever  $r=s+1$  with  $s \geq 1$ . So [13, (34), (36) and (38)] are covered in the same range, by the same count.

Proposition 3.7 and Corollary 3.9 prove Theorem 1.1. They also settle the status of [13, Remark 4.5(3)]: under the standing assumption  $r > s$  its range  $r-s \leq 1$  is exactly  $\nu=1$ , and by Corollary 3.2 and Remark 3.3 its negative assertion holds there only for  $s=0$ .

## 4 The Meijer aperture at $\nu=1$

All arguments and sectors in this section are taken on the universal cover of the punctured  $z/q$ -plane; the sectors below have opening  $2\pi$  and more, and this fixes their branch.

The asymptotic analysis of [13, Section 7] rests on [13, Lemma 7.4], which writes the equivariant central-charge kernel  $\kappa_m$  as a cohomology-valued Meijer  $G$ -function,

$$\kappa_m = e^{-\pi\sqrt{-1}c_1(V')} G_{s,r}^{r,0} \left( \begin{matrix} 1-\sigma_1, \dots, 1-\sigma_s \\ \rho_1, \dots, \rho_r \end{matrix} \middle| t_m \right), \quad t_m := e^{-\pi\sqrt{-1}(2m+s)} \left( \frac{q}{z} \right)^{r-s}. \quad (5)$$

In the notation of [13, Appendix A] the upper and lower parameter rows of this  $G$ -function have  $s$  and  $r$  entries, so the parameter governing the asymptotic expansion is

$$\nu = r - s, \quad (6)$$

which is the discrepancy of the flip. (The appendix of [13] names the index counts  $p$  and  $q$ ; we write  $s$  and  $r$  to keep  $q$  for the extremal variable.)

[13, Theorem A.1] is Barnes' asymptotic expansion [2] in the form given by Meijer [10] and Luke [9]. Its hypothesis is stated there as

$$|\arg t| < (\nu + \epsilon)\pi, \quad \epsilon = 1 \text{ if } \nu > 1, \quad \epsilon = \frac{1}{2} \text{ if } \nu = 1. \quad (7)$$

The  $\epsilon$  of (7) is the whole issue. The expansion printed in [13, Section 7] after [13, Lemma 7.4] is introduced by the clause “For  $r-s > 1$ ”, and the sector recorded with it,

$$\left| \arg\left(\frac{z}{q}\right) + \frac{(2m+s)\pi}{r-s} \right| < \left(1 + \frac{1}{r-s}\right)\pi, \quad (8)$$

which is [13, (64)], is exactly (7) with  $\epsilon=1$  transported through (5).

**Lemma 4.1.** *Let  $\nu=1$ . Applied to (5), [13, Theorem A.1] gives the asymptotic expansion of [13, Proposition 7.5] on*

$$\left| \arg\left(\frac{z}{q}\right) + (2m+s)\pi \right| < \frac{3\pi}{2}. \quad (9)$$

*The sector obtained by substituting  $r-s=1$  into (8) is  $|\arg(z/q) + (2m+s)\pi| < 2\pi$ , wider by  $\pi/2$  on each side.*

*Proof.* From (5),  $\arg t_m = -(2m+s)\pi - \nu \arg(z/q)$ . At  $\nu=1$  the hypothesis (7) reads  $|\arg t_m| < (1 + \frac{1}{2})\pi$ , which is (9). Substituting  $r-s=1$  into (8) gives  $(1+1)\pi = 2\pi$ , the value of  $(\nu + \epsilon)\pi$  at  $\epsilon=1$ .  $\square$

The correction is one-directional: (9) is smaller than the naive specialization of (8), so a repair must show that what remains is still enough. The other sector in play is unaffected.

**Lemma 4.2.** *[13, Proposition 8.2] and its sector*

$$\left| \arg\left(\frac{z}{q}\right) - \frac{(1-s)\pi}{r-s} \right| < \frac{\pi}{2} + \frac{\pi}{r-s}, \quad (10)$$

which is [13, (78)], are valid at  $\nu = 1$ , where (10) reads  $|\arg(z/q) - (1-s)\pi| < \frac{3\pi}{2}$ .

*Proof.* [13, Proposition 8.2] is deduced from [13, (76)], which applies [13, Theorem A.2] to a  $G_{s,r}^{r,1}$  function of the argument  $t = e^{\pi\sqrt{-1}(1-s)}(q/z)^{r-s}$  of [13, (75)]. The hypothesis of [13, Theorem A.2] is  $|\arg t| < (\frac{\nu}{2} + 1)\pi$ ; it involves no  $\epsilon$  and no restriction on  $\nu$ . Transporting it through that substitution gives (10) for every  $\nu \geq 1$ , and the displayed form at  $\nu = 1$ .  $\square$

**Proposition 4.3.** *Let  $r = s + 1$  with  $s \geq 1$  and let  $k \in \{0, 1\}$ , so that the semiorthogonal decomposition [13, Theorem 1.1] has the single  $Z$ -component of index  $m = -k$ . The sectors (9) and (10) meet in the open sector*

$$\max((2k-s)\pi, (1-s)\pi) - \frac{3\pi}{2} < \arg\left(\frac{z}{q}\right) < \min((2k-s)\pi, (1-s)\pi) + \frac{3\pi}{2}, \quad (11)$$

of opening  $2\pi$ . It contains the eigenvalue ray  $\arg(z/q) = (2k-s)\pi$  of [13, Theorem 1.4(1)] and the tame ray  $\arg(z/q) = (1-s)\pi$  of [13, Theorem 1.4(2)].

*Proof.* With  $m = -k$ , sector (9) is the open interval of half-width  $3\pi/2$  about  $(2k-s)\pi$ , and sector (10) is the open interval of half-width  $3\pi/2$  about  $(1-s)\pi$ . For  $k \in \{0, 1\}$  the two centres are at distance  $|2k-1|\pi = \pi$ , so the intervals meet in an interval of length  $3\pi - \pi = 2\pi$ , namely (11). Since  $\pi < 3\pi/2$ , each centre lies in the other interval, so both centres lie in the intersection; they are the two rays named in the statement, because at  $r-s=1$  the rays  $-(2m+s)\pi/(r-s)$  and  $(1-s)\pi/(r-s)$  of [13, Theorem 1.4] are  $(2k-s)\pi$  and  $(1-s)\pi$ .  $\square$

Lemmas 4.1 and 4.2 and Proposition 4.3 prove Theorem 1.2.

*Remark 4.4.* For a blow-up,  $s = 1$ , the numbers are: (9) with  $k = m = 0$  is  $-\frac{5\pi}{2} < \arg(z/q) < \frac{\pi}{2}$ ; (10) is  $|\arg(z/q)| < \frac{3\pi}{2}$ ; and (11) is  $-\frac{3\pi}{2} < \arg(z/q) < \frac{\pi}{2}$ , containing the tame ray 0 and the eigenvalue ray  $-\pi$ . The codimension-two blow-up is  $(r, s) = (2, 1)$ , where  $\lambda_0(q) = -q$  and the exponential factor of the expansion of [13, Proposition 7.5] is  $\exp(-\lambda_0(q)/z) = \exp(q/z)$ .

*Remark 4.5.* The intersection (11) has opening  $2\pi > \pi$ , so it is wider than the level-one threshold beyond which an asymptotic expansion determines its flat section uniquely; see [1]. The oriented comparison of the nonzero-eigenvalue and tame expansions carried out in [13, Section 9], which needs both expansions to be valid simultaneously on a sector containing both rays, therefore remains available at  $\nu = 1$ . In particular the common-sector condition of [13, Remark 1.6], namely  $\frac{1}{4}(r-s-6) < k < \frac{3}{4}(r-s+2)$ , which at  $r-s=1$  reads  $-\frac{5}{4} < k < \frac{9}{4}$  and admits both  $k=0$  and  $k=1$ , holds at  $\nu=1$  for the corrected sector (9) and not only for (8).

## 5 What is repaired, and what is not

**Theorem 5.1.** *Let  $X \dashrightarrow X'$  be a standard flip with  $r = s + 1$  and  $s \geq 1$ . Then [13, Theorems 1.2, 1.4, 9.9 and 9.14] and [13, Corollary 1.5] hold for  $X$ . In particular they hold for the blow-up of a smooth projective variety along a smooth centre of codimension two, the case  $(r, s) = (2, 1)$ .*

*Proof.* Corollary 3.9 gives (4) and [13, Theorem 1.2], and makes [13, (39)–(40)] available, so the modified extremal  $J$ -function used from [13, Section 7] onward is the one the argument assumes. [13, Proposition 7.5], [13, Corollary 7.6] and [13, Proposition 7.8] then hold with (8) replaced by (9), by Lemma 4.1; [13, Propositions 8.2 and 8.3] and [13, Corollary 8.4] hold as printed, by Lemma 4.2.

Once the repaired  $J$ -function formula and the corrected common sector are supplied, [13, Sections 9.1–9.4] impose no further restriction on  $r - s$ , and their internal arguments apply without modification. [13, Proposition 9.1] uses the presentation (4) and the extremal Dubrovin connection; [13, Proposition 9.2] combines it with [13, Proposition 7.8] and [13, Corollary 8.4]; [13, Corollary 9.3] takes a partial nonequivariant limit; [13, Section 9.2] removes the splitting assumption on  $V, V'$ , giving [13, Corollary 9.7]; [13, Section 9.3] passes from the local model to  $X$ , giving [13, Theorem 9.9]; and [13, Section 9.4] compares the Fourier–Mukai transform with the local model, giving [13, Theorem 9.14]. None of these steps constrains  $\nu$  beyond  $r > s$ , but they are not independent of the repair: the deformation argument of [13, Section 9.2] runs through [13, Lemmas 9.4–9.6], whose proofs read (1) as the extremal  $J$ -function. Proposition 3.7 supplies exactly that at  $r - s = 1$ . The two halves of [13, Theorem 1.4] are [13, Theorems 9.9 and 9.14], and [13, Corollary 1.5] is the translation of [13, Theorem 1.4] through the functors [13, (3)]. Proposition 4.3 supplies the sector on which both expansions used in that translation are valid.  $\square$

Three boundaries should be stated explicitly.

First,  $(r, s) = (1, 0)$  is not covered, and it is not another discrepancy-one standard flip in the extremal-line setup. There  $\mathbf{P}(V) = Z$ ,  $X' = \emptyset$ , the fibre is a point, and there is no line class  $[L]$  to restrict the Novikov parameter to. By Remark 3.3 the formal expression (1) has a  $z$ -order-zero term there and is not  $J$ -normalized; that failure marks the boundary of the formula rather than an omitted geometric case. All genuine discrepancy-one standard flips have  $s \geq 1$  and are covered by Theorem 5.1.

Second, everything above lives on the extremal slice. The Novikov parameter is restricted to multiples of the class  $[L]$  of a line in a fibre of  $\mathbf{P}(V)$ , all other Novikov variables are zero, and the parameter is  $\mathbf{t} = t + \ln(q)c_1(T)$  with  $t$  pulled back from  $H^*(Z)$ ; this is the specialization in which [13, Theorems 1.2 and 1.4] are stated, and no statement here concerns the unrestricted quantum product or a comparison across the full Novikov ring.

Third, the two corrections are local to the two hypotheses named in Section 2. For  $\nu > 1$  nothing changes: the argument of [13] applies as printed, with (8) supplied by the  $\epsilon = 1$  branch of [13, Theorem A.1]. Every other ingredient of [13] — the ring homomorphism to the local model, the spectral computation in the abstract presentation, the deformation to the normal cone, the Fourier–Mukai comparison — is used exactly as it stands. No other part of the standard-flip argument is altered: the correction is confined to the  $J$ -normalization of (1) and to the Barnes aperture at  $\nu = 1$ , with the cone membership of Section 3 supplied rather than corrected.

The count in Lemma 3.1 is what makes the discrepancy-one range behave like the higher-discrepancy one. The  $s$  numerator factors  $\sigma_j - H$  at  $m = 0$ , free of  $z$ , are worth exactly  $s$  powers, and at  $\nu = 1$  they are the entire margin: the maximal order  $1 - s - \nu d$  sits at  $-1$  precisely when  $s = 1$  and  $d = 1$ , and rises to 0 the moment  $s$  drops to zero. So the boundary between the identification  $I = J$  and a nontrivial mirror map at  $\nu = 1$  is the presence of at least one factor in  $V'$ , that is  $s \geq 1$ .

## AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted with proof exploration, literature research, exact computation, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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